

SVM training and prediction (linear)

You are working as a data scientist in a fruit company. You would like to develop a model to judge whether a given fruit is orange or grapefruit.

You have a CSV dataset named "fruit_dataset.csv" (see reading materials). The file contains 6 columns:

- name: class labels, can be {0, 1}, where 0 and 1 represent orange and grapefruit, respectively
- diameter: diameter of the fruit in centimeters
- weight: weight of the fruit in grams
- red: average red reading from an RGB scan
- green: average green reading from an RGB scan
- blue: average blue reading from an RGB scan

Train a linear SVM, called 'SVMModel_linear', on the training data.

Use it to predict the labels of the test data and compute the accuracy percentage.

Script

 Save

 Reset

 MATLAB Documentation (https://www.mathworks.com/help/)

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```
1 % Load data
2 data = readtable('fruit_dataset.csv');
3 X = data(:, 2:6);
4 Y = data(:, 'name');
5 % Split into training and test data by the ratio 75% and 25%, respectively.
6 cvObj = cvpartition(size(X, 1), 'HoldOut', 0.75);
7 idx = cvObj.test;
8 X_train = X(idx, :); X_test = X(~idx, :);
9 Y_train = Y(idx, :); Y_test = Y(~idx, :);
10
11 % Train a linear SVM, the data should be standardized, your code:
12 SVMModel_linear = fitcsvm(X_train, Y_train, 'KernelFunction', 'linear', 'Standardize', true);
13
14 % Predict using the linear SM, your code:
15 Y_pred_linear = predict(SVMModel_linear, X_test);
16 accuracy_linear = sum(Y_pred_linear == Y_test) / length(Y_test) * 100;
```

 Run Script



Tests: All Tests Passed

Submit



 Train a linear SVM model that can judge whether a given fruit is orange or grapefruit.

 Predict the linear SVM on the test dataset

 Calculate the prediction accuracy of the linear SVM model. The variable of the prediction accuracy must be named to 'accuracy_linear'.

Output

Code ran without output.